

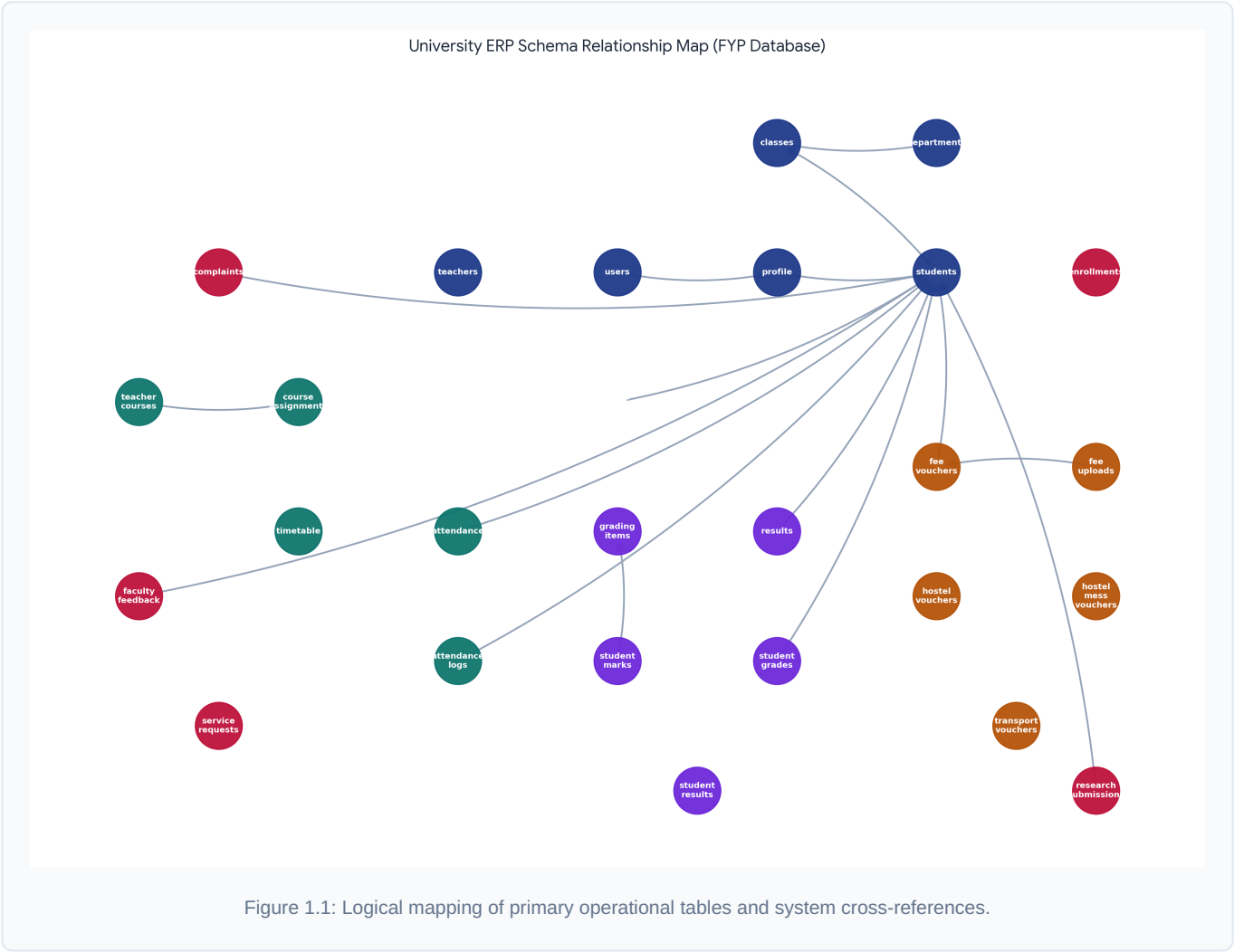
Database Architecture & Relationship Schema

Comprehensive Entity Analysis and Connection Documentation for the University ERP System

1. Executive Summary & Overview

The provided SQL relational schema defines a comprehensive University ERP / Student Information System environment. The architecture contains exactly 31 structural tables organized into logical clusters addressing student records, academic progress, sessional grading, financial management, accommodation services, and user authentication layers.

2. System Entity-Relationship Diagram



3. Functional Module Breakdowns & Foreign Connections

The system uses a mix of explicit relational constraints (Foreign Keys) and programmatic logical keys (e.g., matching *registration_no* or *course_code* strings) to handle standard data lookups:

A. Student Identity & Base Administration Module

The tables **users**, **profile**, and **students** manage the primary user identity records. The **profile** table acts as a comprehensive master ledger detailing personal, family, and past academic metrics (FSc and SSC marks). The **classes** table connects directly back to **departments** via an explicit Foreign Key constraint (*dept_id* → *departments.id*).

B. Academic & Course Delivery Module

course_assignments maps instructors to courses and credit hours for specific semesters. The **teacher_courses** table forms a strict relational dependency mapping back to **course_assignments** via a cascading delete foreign key constraint (*course_id* → *course_assignments.id ON DELETE CASCADE*). Student status is mapped using **attendance** tracking logs and localized daily entry rows in **attendance_logs** linked by course code keys.

C. Financial Accounting & Voucher Management Module

This module controls student billing records. **fee_vouchers** serves as the central ledger for tuition, examination, and laboratory fees. When a payment proof file is submitted, it registers into **fee_uploads**, which has a rigid structural reference pointing directly back to the original ledger line (*voucher_id* → *fee_vouchers.id*). Auxiliary bills are isolated inside **hostel_vouchers**, **hostel_mess_vouchers**, and **transport_vouchers**.

D. Examination and Grading Evaluation Module

Sessional distributions (Quizzes, Assignments, Midterms, and Finals) are configured inside **grading_items**. Individual student outcomes map into **student_marks** using an explicit foreign relation pointing to the target evaluation type (*item_id* → *grading_items.id*). Summarized semester grade performance forms lines inside **student_grades** linked to the primary student identity record.

4. Summary of Hard Schema Constraints (Foreign Keys)

Source Table	Foreign Key Column	Target Referenced Table	Target Primary Column	Behavior Rule
classes	dept_id	departments	id	RESTRICT / Default
exam_datesheet	registration_no	students	registration_no	RESTRICT / Default
fee_uploads	voucher_id	fee_vouchers	id	RESTRICT / Default
student_grades	registration_no	students	registration_no	RESTRICT / Default
student_marks	item_id	grading_items	id	RESTRICT / Default

Source Table	Foreign Key Column	Target Referenced Table	Target Primary Column	Behavior Rule
teacher_courses	course_id	course_assignments	id	ON DELETE CASCADE